

A Brief Discussion of Previous Math Chatbot Research and Projects

This paper will discuss 3 academic papers and 3 existing math chatbots.

Academic Papers

Paper #1:

[\(PDF\) Exploration of Chatbots in Mathematics Education for Innovative Learning Process](#)

This paper mentions that students are concerned about correctness of answers from chatbots. My tool will eliminate this problem by just getting actual answers directly inputted from the teachers that setup the chatbot (math teachers will setup problems-and-answers in JSON files). My tool will use the ChatGPT API to check whether a student answer is equivalent to a correct answer (but not necessarily verbatim in the same format/phrasing, e.g. $\frac{4}{5}$ and 0.8 are actually the same) .

Another issue mentioned is student dependency on chatbots, on page 8. A way to mitigate chatbot dependency is to require a certain proportion of hours of active math discussion with humans. In addition, basic assessments using testing should not be reduced, but maintained as a standard check for student growth on an individual basis and also across student populations.

It would be useful if students treated math chatbots as a means to increase their performance in human-to-human math discussions and in measured assessments. This would be analogous to a human player practicing against a computer chess program before playing chess against other humans in competitions. Or it could be analogous to students practicing a musical instrument before a performance in front of a human audience.

The paper above specifically mentions the “CiCi” chatbot which I had never heard of before, and that this math chatbot was more often used by students than any other bot.

Paper #2:

(A briefer academic paper.)

<https://amte.net/sites/amte.net/files/Connections%20%28Chandler%29.pdf>

This paper brings up some similar ideas to paper #1, and provides some specific examples of AI “hallucinations” (providing wrong answers). Again, my AI chatbot will

actually get answers from the teacher who also inputs the problems, and will use interactions with the ChatGPT API to check possible equivalence between correct answers.

Paper #3:

(Another brief academic paper.)

<https://amte.net/sites/amte.net/files/Connections%20%28Beauchamp%29.pdf>

This paper brings up the topic of generative AI currently showing limited capability of customizing math word problems for students of different cultures/nationalities. An example is brought up where the MagicSchool AI tool responded to a teacher's request to customize an amusement park word problem, for a Bolivian student, by literally only changing its title to say "A Bolivian Amusement Park" (that is to say just adding the adjective "Bolivian" before "Amusement Park").

This topic will not be addressed directly by my math chatbot. Instead, because my AI chatbot will simply be a wrapper decorator to the OpenAI API, others can in turn wrap-and-decorate my chatbot in foreign-language and cultural interface layers. Please see the page 1 diagram in my capstone project proposal:

https://docs.google.com/document/d/1bcU3x3DSFJWHhJOrmhUvGmPu4p_D90XGpKlcL43_S1A/edit?usp=sharing

Foreign-language wrappers can be a particularly exciting use of existing AI technology to aid English-as-a-Second-Language (ESL) students in learning math.

My AI chatbot project is meant to be *additive* toward currently-existing math capabilities already in ChatGPT accessible through the API, while allowing in turn *other tools to decorate and further add functionality to it*. Future tools can wrap my chatbot's math capabilities with their own additional software features.

Existing Math Chatbots

Unlike my math chatbot, the existing math chatbot projects that I have seen are based upon interactions with the Gemini AI tool (from Google) rather than ChatGPT (from OpenAI). But otherwise there is some degree of similarity between my project math chatbot and these existing tools, both in functionality and structure.

The main 2 customer-level differences between these other math chatbots and my project are that:

1. My math chatbot will specifically concern arithmetic number talks (mental math conversations).
2. My math chatbot will input its math problems, answers, and related math properties from actual human teachers (using JSON, or possibly YAML input files). An example teacher-inputted setup for the problem $996 / 4$ would be:

```
[ ...
{ problem: "1000 / 4"
  answer: 250
  related_problems: [ '4 * 250' ]
},
{ problem: "996 / 4"
  answer: 249
  related_problems: [ '1000 / 4' ]
  related_properties: [ { property: 'distributive'
                        usage: '996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) *
(1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249'
                      }
                    ]
  common_errors: [ { error: '996 / 4 = 6'
                    Reasoning: '9 + 9 + 6 = 24, 24 / 4 = 6' }
                  ]
}
]
```

The main characteristics of my math chatbot design will be:

1. The chatbot software will be a decorator wrapper to the Open AI API, pre-processing inputs and post-processing outputs.
2. The chatbot software will simply take the parts of a problem and solution, and place those parts into pre-processed inputs to ChatGPT without much regard for grammar or structure, allowing the ChatGPT tool itself to sort out such details. My math chatbot will also use ChatGPT to determine if different-looking answers are equivalent.
3. My chatbot software will implement rudimentary cross-attention between teacher inputs, student inputs, and ChatGPT API inputs and outputs. This cross-attention between phrases will only determine a very rough correlation where overlapping-content phrases have encoded dot products closer to one, whereas less-related phrases have encoded dot products closer to zero. This

will allow my math chatbot to better choose for relevance between multiple possible responses that the ChatGPT API provides.

ChatGPT itself already has some built-in math capabilities. My math chatbot can leverage these capabilities for two main reasons:

1. My mathbot can use ChatGPT to check whether a child's math answer is incorrect or just is just a correct answer expressed in a different form than expected (sometimes equivalent correct answers look different). For instance, my chatbot can ask ChatGPT: "A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = 249$. Just need to verify that the child is incorrect and that the correct answer is indeed 249." ChatGPT answers: Yes. The child is **incorrect**.
2. My mathbot can then ask GPT to provide the child a path to better thinking: My math chatbot will ask ChatGPT: "A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child."

.. and ChatGPT provides two responses:

1. "How can you rewrite 996 as 1000-4 and use the distributive property to divide each part by 4, then combine the two results?"
2. "If $9+9+6=24$ and $24\div4=6$, how could you use multiplication to check whether 6 is actually the quotient of $996\div4$, and then use $996=(1000-4)$ to find the correct quotient?"

... this allows my math chatbot to select among the two responses (my chatbot would select the 1st answer because it includes the distributive property that the previous question fed into ChatGPT mentioned).

Part of the job of my project math chatbot will be to choose the best answer that ChatGPT offers based upon a simple attention scheme, seeing if encoded phrases have high dot products (dot products closer to 1 rather than to 0).

Existing Math Chatbot #1, “MathMate:”

The following GitHub website contains something pretty close to what I want to do with my Capstone project:

[GitHub - RidaHajjAli/MathMate: An Intelligent Chatbot with Advanced Mathematical Skills · GitHub](#)

The “MathMate” tool specifically routes user interactions to “sympy” (a Python library for symbolic mathematics) for math discussion and to Gemini for non-math discussion.

The MathMate tool, leveraging Gemini, already produces reasonable answers to arithmetic questions. For example...

User input: A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = 249$. Just need to verify that the child is incorrect and that the correct answer is indeed 249.

MathMate: Yes, that is correct. The child is incorrect, and the correct answer is indeed 249.

Verification:

- $249 \times 4 = 996$

(Note: The child likely added the digits together $9 + 9 + 6 = 24$ and then divided $24 \div 4 = 6$.)

The screenshot shows a web browser at localhost:8501 displaying the "Gemini Chatbot with SymPy Calculator" interface. On the left is a sidebar with a "New Chat" button and a "Conversation History" section showing three previous exchanges. The main chat area has a title "Gemini Chatbot with SymPy Calculator" and a subtitle "Ask general questions or perform calculations (e.g., $\sqrt{16}$ or $\sin(\pi/2)$)." The chat history shows three messages: 1. A red bubble from the user: "A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = 249$. Just need to verify that the child is incorrect and that the correct answer is indeed 249." 2. An orange bubble from the chatbot: "Yes, that is correct. The child is incorrect, and the correct answer is indeed 249. Verification: • $249 \times 4 = 996$ (Note: The child likely added the digits together $9 + 9 + 6 = 24$ and then divided $24 \div 4 = 6$.)" 3. A red bubble from the user: "A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child." At the bottom is a text input field with the placeholder "Your message or calculation..." and a send button.

User input: A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child.

MathMate:

Here are 2 productive questions you can ask the child to guide their thinking toward the correct reasoning without giving away the final answer:

Option 1: Using a Benchmark / Friendly Number (Distributive Thinking)

“996 is very close to the friendly number 1,000. If we know that 1,000 divided by 4 is 250, how can using 1,000 help us figure out what 996 divided by 4 would be?”

- Why this works: It introduces the correct reasoning of using $1000 / 4$ as a starting point, prompting the child to realize the answer must be close to 250 rather than a small single-digit number like 6.

The screenshot shows a web application interface for MathMate. The browser address bar shows 'localhost:8501'. The interface has a sidebar on the left with a 'New Chat' button and a 'Conversation History' section showing 'Last 10 exchanges (newest first):' with three entries: 'Exchange 3', 'Exchange 2', and 'Exchange 1'. The main chat area on the right shows a conversation. At the top, a user input is shown: '249 x 4 = 996' with a note: '(Note: The child likely added the digits together 9 + 9 + 6 = 24 and then divided 24 ÷ 4 = 6.)'. Below this, a system message (orange icon) says: 'A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child.' Below this, another system message (orange icon) says: 'Please input a mathematical problem for me to solve.' Then, a third system message (orange icon) says: 'Here are 2 productive questions you can ask the child to guide their thinking toward the correct reasoning without giving away the final answer:'. This is followed by the section header 'Option 1: Using a Benchmark / Friendly Number (Distributive Thinking)'. Below this, a quote says: '“996 is very close to the friendly number 1,000. If we know that 1,000 divided by 4 is 250, how can using 1,000 help us figure out what 996 divided by 4 would be?”'. Finally, a bullet point says: '• Why this works: It introduces the correct reasoning of using $1000 / 4$ as a starting point, prompting the child to realize the answer must be close to 250 rather than a small single-digit number like 6.' At the bottom, there is a text input field with the placeholder 'Your message or calculation...' and a 'Deploy' button in the top right corner.

Option 2: Place Value & Estimation (Magnitude Check)

“If you have almost 1,000 items to split equally into 4 equal piles, about how many items should be in each pile? How can we divide the actual value of 996 instead of adding up the single digits?”

- Why this works: It helps the child spot the flaw in their digit-summing method ($9 + 9 + 6$) by redirecting their attention to the actual place value/magnitude of 996.

Some interesting sidenotes:

Interestingly, the “MathMate” chatbot actually offers safety filtering against harassment, hate speech, sexually explicit content, and dangerous content. This is a subject that I had never really thought about, and it is a feature that would best be incorporated also into my project chatbot if I have the time and bandwidth to put in these filtering features. More likely, I will leave this sensitive subject to be handled by other, later decorator wrappers around my math chatbot, just as I will leave foreign-language translation wrappers as later work.

Existing Math Chatbot #2, “ai-math-chatbot:”

[GitHub - EvanGks/ai-math-chatbot: AI-powered math chatbot with Gemini, Whisper, file upload, and voice input · GitHub](#)

This chatbot similarly uses Gemini, and similarly provides useful answers to example math questions...

User input: A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = 249$. Just need to verify that the child is incorrect and that the correct answer is indeed 249.

ai-math-chatbot:

Verification of the Calculation

To determine if the child is incorrect, we must verify the exact mathematical result of the division problem.

We need to verify two things:

1. Whether $996/4$ is indeed equal to 249.
2. Whether the child's answer of 6 is incorrect.

We can calculate the quotient by decomposing the dividend using subtraction:

$$996 = 1000 - 4$$

Now we divide this expression by 4:

$$996/4 = (1000 - 4)/4$$

Using the distributive property of division over subtraction, we rewrite the expression:

$$(1000 - 4)/4 = (1000/4) - (4/4)$$

We simplify each term: $250 - 1 = 249$

=> The correct quotient of $996/4$ is indeed 249.

Verification of the Child's Error

The child believes the result is: 6

We compare the child's proposed answer with the correct verified quotient:

$$6 \neq 249$$

=> The child's answer of 6 is incorrect, and the correct answer is indeed 249.

Final Answer: 249

Would you like to look at some different methods for explaining this division to the child, such as using money (like sharing 996 among 4 people) to help make the scale of the numbers clearer to them?

User input : A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child.

Ai-math-chatbot:

To help the child recognize their mistake and guide them toward correct mathematical reasoning, we want to address the core of their misconception: place value and scale/estimation.

The child is likely adding the individual digits of the dividend:

$$9+9+6=24$$

And then dividing that sum by the divisor:

$$24/4=6$$

This method treats the digits as single units, overlooking that:

$$996=900+90+6$$

Here are two productive questions designed to help the child discover the correct reasoning.

Productive Question 1: Focus on Estimation and Scale

This question uses a friendly benchmark number (1000) to help the child realize that their answer of 6 is far too small.

Here is the first question: "We know that the number 996 is very close to 1000. If we have 1000 pennies and share them equally among 4 friends, each friend gets 250 pennies because $1000/4=250$. Since 996 is almost the same as 1000, how does that help us see whether our share of $996/4$ should be closer to 250 or closer to 6?"

Productive Question 2: Focus on Place Value Decomposition

This question prompts the child to think about the actual value of each digit, guiding them toward the distributive property.

Here is the second question: "In your calculation, you added $9+9+6$. But in the number 996, is that first '9' worth just 9 ones, or is it worth 900? If we write 996 as $900+90+6$, how would we share each of those parts among 4 groups instead of just adding the single digits?"

Final Answer:

Question 1: "Since 996 is close to 1000, and $1000/4=250$, should the answer be close to 250 or 6?"

Question 2: "Does the first 9 in 996 represent 9 or 900? How does that change how we divide?"

Which of these two questions do you think would resonate more with the child's current learning style? Would you like to explore how to use visual aids, like play money or base-10 blocks, to back up these questions?{"text": ""}

Existing Math Chatbot #3, “Math-Solving-Assistant:”

[GitHub - duongtruongbinh/Math-Solving-Assistant: A chatbot website utilizing Gemini's API for solving math problems, chatting, and image analysis · GitHub](#)

The “Math-Solving-Assistant” tool is similar to the other two math chatbots (above) in that it also leverages Gemini. One difference with this 3rd math chatbot is that it has incorporated fine-tuning of the Gemini model for math, using the dataset at:

<https://huggingface.co/datasets/meta-math/MetaMathQA-40K>

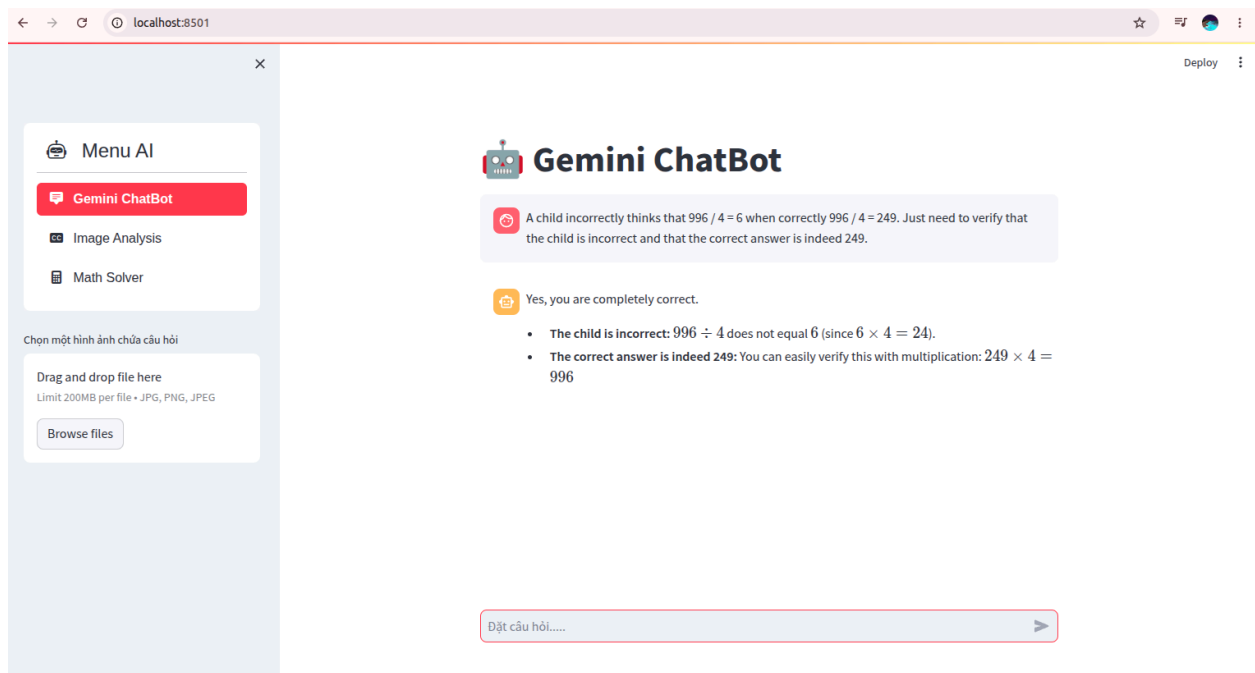
Math-Solving-Assistant provided useful answers to the example math questions, similar to the other tools...

User input: A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = 249$. Just need to verify that the child is incorrect and that the correct answer is indeed 249.

Math-Solving-Assistant:

Yes, you are completely correct.

- The child is incorrect: $996 \div 4$ does not equal 6 (since $6 \times 4 = 24$).
- The correct answer is indeed 249: You can easily verify this with multiplication: $249 \times 4 = 996$



User input: A child incorrectly thinks that $996 / 4 = 6$ when correctly $996 / 4 = (1000 - 4) / 4 = (1000 + (-4)) * (1/4) = (1000 * (1/4)) + ((-4) * (1/4)) = (1000 / 4) + (-4/4) = 250 - 1 = 249$, using the distributive property. Ask a question that asks the child a productive question (not an answer) containing correct reasoning. The child might have incorrectly reasoned that $996 / 4 = 6$ because $9 + 9 + 6 = 24$ and $24 / 4 = 6$. Provide 2 possible productive questions to ask the child.

Math-Solving-Assistant:

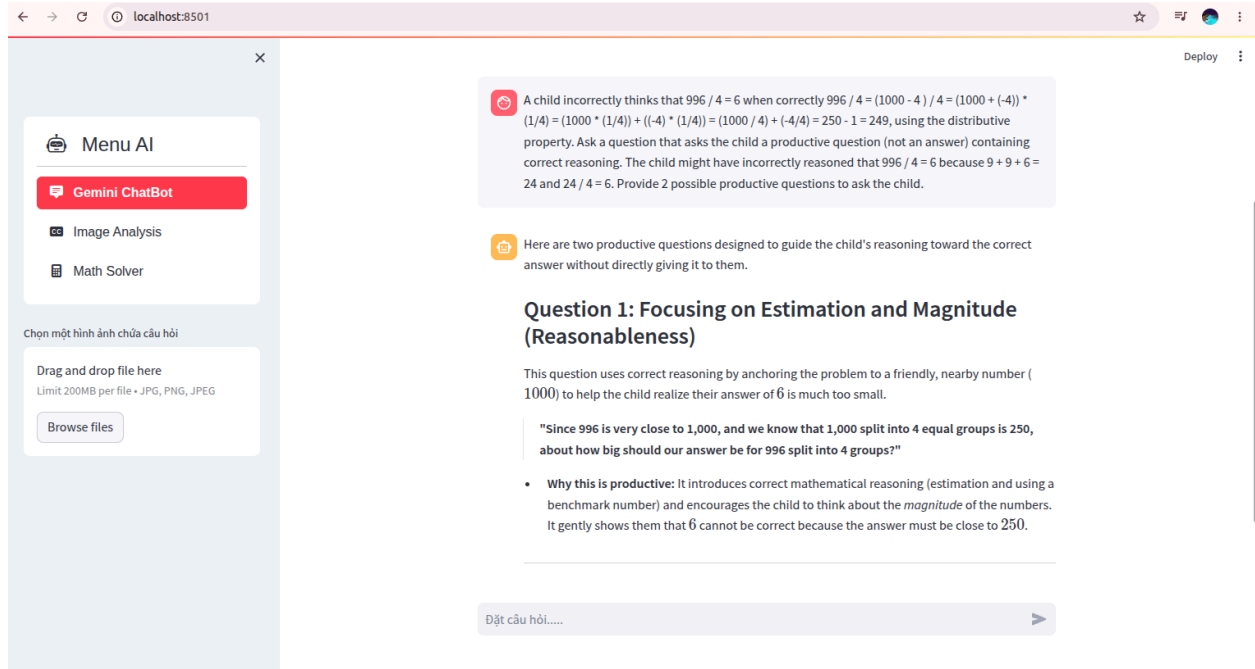
Here are two productive questions designed to guide the child's reasoning toward the correct answer without directly giving it to them.

Question 1: Focusing on Estimation and Magnitude (Reasonableness)

This question uses correct reasoning by anchoring the problem to a friendly, nearby number (1000) to help the child realize their answer of 6 is much too small.

"Since 996 is very close to 1,000, and we know that 1,000 split into 4 equal groups is 250, about how big should our answer be for 996 split into 4 groups?"

- Why this is productive: It introduces correct mathematical reasoning (estimation and using a benchmark number) and encourages the child to think about the *magnitude* of the numbers. It gently shows them that 6 cannot be correct because the answer must be close to 250.



Question 2: Focusing on Place Value and the Distributive Property

This question directly addresses the likely misconception (treating the digits 9, 9, and 6 as single units) by guiding them to look at the actual value of the number 996.

"When we do $9+9+6=24$, we are treating those digits like they are just ones. But 996 is actually just 4 less than 1,000. If we split 1,000 into 4 groups (which is 250), how can we adjust that to split 996 into 4 groups?"

- Why this is productive: It validates the child's math ($24 \div 4 = 6$) but points out the place value error (the digits aren't just ones). It then models the correct reasoning of breaking the number down ($1000 - 4$) and asks the child to think about how to subtract the difference to find the final result.

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Deploy ⋮

✕

Menu AI

Gemini ChatBot

Image Analysis

Math Solver

Chọn một hình ảnh chứa câu hỏi

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1000) to help the child realize their answer of 6 is much too small.

"Since 996 is very close to 1,000, and we know that 1,000 split into 4 equal groups is 250, about how big should our answer be for 996 split into 4 groups?"

- Why this is productive: It introduces correct mathematical reasoning (estimation and using a benchmark number) and encourages the child to think about the *magnitude* of the numbers. It gently shows them that 6 cannot be correct because the answer must be close to 250.

Question 2: Focusing on Place Value and the Distributive Property

This question directly addresses the likely misconception (treating the digits 9, 9, and 6 as single units) by guiding them to look at the actual value of the number 996.

"When we do $9 + 9 + 6 = 24$, we are treating those digits like they are just ones. But 996 is actually just 4 less than 1,000. If we split 1,000 into 4 groups (which is 250), how can we adjust that to split 996 into 4 groups?"

- Why this is productive: It validates the child's math ($24 \div 4 = 6$) but points out the place value error (the digits aren't just ones). It then models the correct reasoning of breaking the number down ($1000 - 4$) and asks the child to think about how to subtract the difference to find the final result.

Đặt câu hỏi..... ➤

Extra links:

[Math AI](#)

Rubric on grading summary of previous results:

<https://docs.google.com/document/d/1Ut-DuO5ITQrvigO4JWig8qCuqtSF-jiC/edit>